

Contents

The Accumulating-Context Agent	1
Thesis	1
1. Reprice the Unit of Value	1
2. What Months Add	2
3. Custody Across Engine Changes	2
4. Memory Needs Two Times, and Ledgers Must Tell the Truth	2
5. Context Hygiene Is an Engineering Discipline	3
6. Evidence That Compounds	4
AI-involvement statement	4
References	4

The Accumulating-Context Agent

Authors: Prof. Dr. Mustafa Melikoğlu · Yağız Deniz Altınbaş · Tayfun Tanrıöver

Thesis

The unit of value in agent systems is not the model call — it is the **accumulated context**: an identity that has read its own ledgers for months can draw on decisions and corrections a fresh prompt would have to reconstruct from scratch. Engines rent intelligence; the accumulation is owned. This is why identity lives on disk, why memory is bi-temporal, why ledgers must not lie, and why we treat context-hygiene as an engineering discipline with brakes, anchors, and pickup chains.

1. Reprice the Unit of Value

A model call is easy to count. It arrives with timestamped output and a bill. Repeated work makes that accounting look thin. Before a useful answer appears, earlier sessions may already have settled the terminology or recorded why an attractive framing failed. The call performs the present inference while drawing on work whose cost sits elsewhere.

Article spines in our publication program draw on dated working records. In the records reviewed at the publication gate, at least two spines preserve an earlier correction or narrowed claim alongside the reason for the change. This makes inheritance inspectable. It does not establish a publishable-spine count, a common observation period, drafting speed, or a causal advantage from accumulated context.

Earlier text alone tells us little. A later draft may carry a correction with its rationale, or preserve a question that was deliberately left open. Those relations are part of the work. The record can also preserve an error.

An unpublished working synthesis authored by an AI agent and dated 13 August 2026 records the provenance of this framing; it is cited as a source record, not as independent evidence (Marcus, 2026). A separately framed public version is archived as *Core Synthesis: Agent Orchestration and Harness Sovereignty — an Internal Memo, Published As-Is* (concept DOI: <https://doi.org/10.5281/zenodo.22024504>). Because the public artifact descends from the working synthesis, it is also provenance rather than independent confirmation of this essay's claims.

2. What Months Add

Months give retained material an order. Decisions acquire reasons, and corrections point back to the claims they supersede. A long prompt can reproduce the sentences and still lose those relations.

Identity is practical in this account. It locates responsibility across sessions: the agent meets its earlier commitments and sees where its judgment changed. We are describing working continuity that can be inspected, without making a claim about consciousness.

The present record does not support a speed comparison between a fresh start and a resumption. It lacks a paired baseline, matched tasks, a common observation window, consistent endpoints, and predeclared exclusions. The narrower point is inspectable: a resumed agent can consult recorded decisions and corrections that a fresh start would have to reconstruct. Task familiarity or a better brief could account for any faster start.

The narch Manifesto and the Invariants Compact publicly describes persistent identity as an operating commitment (Melikoğlu, Altınbaş, & Tanrıöver, 2026; concept DOI: <https://doi.org/10.5281/zenodo.22019441>). That companion record establishes the stated commitment, not its comparative effectiveness. The relevant material here is whatever record was actually available when work resumed.

3. Custody Across Engine Changes

Model access comes under terms the user does not control. An owned working record can remain available when those terms change or a different engine takes over. The replacement may read the same history and reach a different judgment.

The retained history gives reviewers something concrete against which to notice the shift.

Handoffs expose the requirement in ordinary work. When a session ends midstream, the next one must recover what still governs. Open questions need to be recognizable, since inherited notes are not firsthand observation.

One documented handoff was sealed at 00:25:34 on 10 August 2026 and consumed at 00:43:08. The receiving session verified all 9/9 items in the transfer package and explicitly inherited open work and known limits. The interval was 17 minutes 34 seconds. This single case supports continuity of a bounded record across one interruption; it does not establish lossless transfer or a performance advantage.

The public *Core Synthesis* record describes an engine-independent design in which identity, memory, and doctrine live on disk while the driving model is replaceable (Melikoğlu, Altınbaş, & Tanrıöver, 2026; concept DOI: <https://doi.org/10.5281/zenodo.22024504>). It is a published statement of the architecture, not evidence that any particular handoff is lossless. In practice, the owner still has to review old entries and correct them.

4. Memory Needs Two Times, and Ledgers Must Tell the Truth

Later knowledge often changes how an earlier event is understood. If the revised account reads as though everyone knew it from the beginning, the record becomes tidier and less useful. It no longer explains why the earlier choice made sense.

Bi-temporal memory here names a discipline of supersession and provenance. It keeps the event time distinct from when its account became known, then places a correction when it became available. This is not a claim to a full dual-axis system with independently managed valid and transaction time. The narrower formulation is informed by Johnston and Weis's distinction between valid time and transaction time in their treatment of temporal databases (2010; <https://doi.org/10.1016/C2009-0-20326-6>).

Consider an ordinary editorial handoff. A note says a quotation has been checked, so the editor sends the resumed passage to copy review unchanged. Later, the source message turns up: the check covered the wording, not the attribution. The ledger should preserve the note that governed the handoff and mark when the narrower reading became known.

One documented handoff supplies a sanitized example. On 18 August 2026, an update passed its readback while a discrepancy was provisionally attributed to the file-service surface. A ruling on 19 August traced the discrepancy to our own synchronization bridge and explicitly preserved the earlier PASS while correcting its attribution. The first record shows what governed on 18 August; the correction shows what became known on 19 August. The sequence demonstrates supersession without pretending that the later explanation was available from the start.

Across a handoff, familiar wording can lend weak evidence an authority it never had. The recipient was not present for the discussion. The record has to leave the correction visible alongside the reason it displaced the earlier account.

Ghost Debt: When “Done” Never Leaves the Ledger (and “Planned” Never Enters It) examines the narrower problem of status drifting away from evidence (Melikoğlu, Altınbaş, & Tanrıöver, 2026; concept DOI: <https://doi.org/10.5281/zenodo.22019942>). The present essay uses that problem at handoff, when a status entry and its support no longer match.

5. Context Hygiene Is an Engineering Discipline

Context hygiene is maintenance for finite attention. No pickup begins by rereading everything. Someone has to judge what still governs and what provenance may matter. Often the work is no more glamorous than checking an inherited statement before using it.

When inherited context lacks evidence or conflicts with a standing commitment, the work needs a brake. Its value is established in use, especially when continuing would be convenient. *Written ≠ Wired: Hunting Fake-Green in Agent Systems* reports measured fail-open protections — including a brake — and their counterfactual closure in $5 + 3 = 8$ tests, and argues that evidence must be read from the live surface (Melikoğlu, Altınbaş, & Tanrıöver, 2026; concept DOI: <https://doi.org/10.5281/zenodo.22018790>). That precedent supports the narrower rule that a written brake needs an observed test; it does not measure the effectiveness of the context practice described here.

Anchors address quieter drift. A current reference keeps work oriented while old states remain available for correction. At resumption, a pickup chain carries what the next session needs and marks the gaps. Deeper provenance can wait until the stakes call for it. *Counterfactual or Clobber: the Sync Bridge That Silently Ate an Agent’s Memory* documents a stale synchronization bridge overwriting an agent’s memory and the counterfactual acceptance test adopted for that bridge (Melikoğlu, Altınbaş, & Tanrıöver, 2026; concept DOI: <https://doi.org/10.5281/zenodo.22019570>). It supports testing a continuity safeguard against the failure it is meant to prevent, not a general claim that continuity is assured.

Routine review is selective. The current reference may be enough for a low-stakes wording choice. A claim that could alter a standing commitment warrants closer attention to its provenance. This judgment stays open to revision as the consequences become clearer.

The records contain successful resumptions alongside missed deliveries, stale assumptions, and incomplete inheritances. They do not define a comparable cold-start sample or an outcome measure, so no reduction in reorientation time can be claimed. At most, the artifacts show that a maintained pickup chain can expose current commitments and known gaps. Editorial repetition could still shorten startup, especially as the brief

became clearer; that alternative belongs in any later test.

Some entries will remain unverified because their provenance was never kept.

6. Evidence That Compounds

The publication program leaves an inspectable chain from a dated synthesis through article spines to finished sibling pieces. That chain lets a reviewer compare what a piece inherited with what it retained or corrected. It does not establish drafting time, a causal advantage, or a cumulative performance measure.

Pickup and handoff observations may support that account once their comparison conditions are known. For now, they remain secondary.

Published sibling pieces provide another trace across separate acts of writing. A companion empirical article reports a four-experiment, eight-task A/B comparison as one bounded observation of disk-anchored identity portability across a bundled, confounded arm contrast; it neither isolates a causal foundation-model effect nor establishes equivalence (Melikoğlu, Altınbaş, & Tanrıöver, 2026; concept DOI: <https://doi.org/10.5281/zenodo.22086013>). It is a public trace, not evidence that accumulated context caused coherence among the pieces here. Ordinary editorial repetition acting on a strong record could produce that coherence too. A reviewer needs to recover what a piece inherited before assigning much weight to it.

Sometimes a later piece carries the correction but not the note that explains it, leaving the reviewer unable to judge the support for the current wording.

When that basis cannot be recovered, the owner decides whether the draft can proceed with a narrower statement.

In practice, someone has to set aside time for unfinished bookkeeping. A current status gets compared with its support; a missing source stays marked as missing. Sometimes the day's result is an unresolved note left where the next reviewer can see it.

AI-involvement statement

AI systems assisted drafting, translation, source triage, ledger re-derivation, and consistency checks. Human authors retained responsibility for the thesis, evidence boundaries, citation decisions, and final publication approval.

References

- Marcus. (2026, August 13). *MARCUS CORE SYNTHESIS — Agent Orchestration & Harness Sovereignty* [Unpublished internal working synthesis; provenance only].
- Melikoğlu, M., Altınbaş, Y.D., & Tanrıöver, T. (2026, August 20). The narch Manifesto and the Invariants Compact. *Zenodo*. <https://doi.org/10.5281/zenodo.22019441>
- Melikoğlu, M., Altınbaş, Y.D., & Tanrıöver, T. (2026, August 20). Core Synthesis: Agent Orchestration and Harness Sovereignty — an Internal Memo, Published As-Is. *Zenodo*. <https://doi.org/10.5281/zenodo.22024504>
- Johnston, T., & Weis, R. (2010). *Managing Time in Relational Databases: How to Design, Update and Query Temporal Data*. Morgan Kaufmann/Elsevier. <https://doi.org/10.1016/C2009-0-20326-6>
- Melikoğlu, M., Altınbaş, Y.D., & Tanrıöver, T. (2026, August 20). Ghost Debt: When “Done” Never Leaves the Ledger (and “Planned” Never Enters It). *Zenodo*. <https://doi.org/10.5281/zenodo.2201942>

- Melikoğlu, M., Altınbaş, Y.D., & Tanrıöver, T. (2026, August 20). Written $\neq$ Wired: Hunting Fake-Green in Agent Systems. *Zenodo*. <https://doi.org/10.5281/zenodo.22018790>
- Melikoğlu, M., Altınbaş, Y.D., & Tanrıöver, T. (2026, August 20). Counterfactual or Clobber: the Sync Bridge That Silently Ate an Agent's Memory. *Zenodo*. <https://doi.org/10.5281/zenodo.22019570>
- Melikoğlu, M., Altınbaş, Y.D., & Tanrıöver, T. (2026, August 24). Portability of a Disk-Anchored Agent Identity Across Foundation Models: A Blind, Internally Pre-Sealed A/B Comparison. *Zenodo*. <https://doi.org/10.5281/zenodo.22086013>